torch.logsumexp#

https://pytorch.org/docs/stable/generated/torch.logsumexp.html#torch.logsume

Description:

Returns the log of summed exponentials of each row of the input tensor in the given dimension dim. The computation is numerically stabilized. For summation index jjj given by dim and other indices iii, the result is If keepdim is True, the output tensor is of the same size as input except in the dimension(s) dim where it is of size 1. Otherwise, dim is squeezed (see torch.squeeze()), resulting in the output tensor having 1 (or len(dim)) fewer dimension(s). input (Tensor) – the input tensor. dim (int or tuple of ints) – the dimension or dimensions to reduce.

Function Signature

```
torch.logsumexp(input, dim, keepdim=False, *, out=None)#
Parameters
Keyword Arguments
```

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.randn(3, 3)
>>> torch.logsumexp(a, 1)
tensor([1.4907, 1.0593, 1.5696])
>>> torch.dist(torch.logsumexp(a, 1),
torch.log(torch.sum(torch.exp(a), 1)))
tensor(1.6859e-07)
```

Detailed Documentation

Documentation

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For summation index jjj given by dim and other indices iii, the result is

If keepdim is True, the output tensor is of the same size as input except in the dimension(s) dim where it is of size 1. Otherwise, dim is squeezed (see torch.squeeze()), resulting in the output tensor having 1 (or len(dim)) fewer dimension(s).

input (Tensor) – the input tensor.

dim (int or tuple of ints) – the dimension or dimensions to reduce.

keepdim (bool, optional) – whether the output tensor has dim retained or not. Default: False.

out (Tensor, optional) – the output tensor.